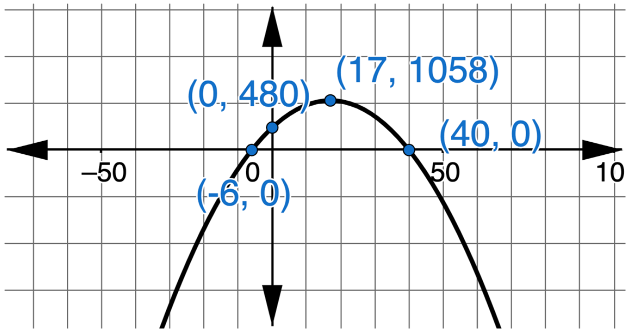
8.8.3 Wrap-Up: Ticket Out the Door

1. A statistician gathers data over several years, starting in 1995, on the average number of trips a household makes to run errands per year. The statistician uses the data to find an equation that models the average annual number of trips for errands per household for the years since 1995. The statistician determined the function $E(x) = -2x^2 + 68x + 480$ models the situation.

   The graph of the function is shown.

   a. Interpret $E(17)$. Include units.

   b. The x-intercepts are labeled on the graph. Determine how reasonable the x-intercepts are by explaining what they each mean in terms of when the average annual number of trips for errands per household will reach 0.

Unit 8, Lesson 9: Using Key Features to Solve Problems

Benchmarks

MA.912.AR.3.6 Given an expression or equation representing a quadratic function, determine the vertex and zeros and interpret them in terms of a real-world context.

MA.912.AR.3.8 Solve and graph mathematical and real-world problems that are modeled with quadratic functions. Interpret key features and determine constraints in terms of the context.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.

Learning Targets

- ☐ I can solve real-world problems modeling quadratic relationships by determining the necessary key feature, and the form of the equation that reveals it.
- ☐ I can interpret key features of quadratic relationships in a real-world context.

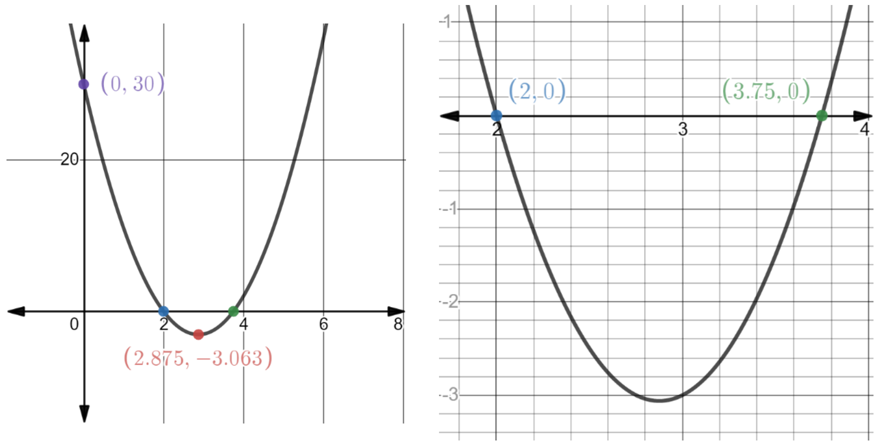
8.9.1 Warm-Up: Key Features and Forms

Several investors provided funds for a startup company to begin production of a new product. The company's revenue, R, in thousands of dollars, for the first year is modeled by the function $R(x) = 4x^2 - 23x + 30$ where x is the number of months since production started.

1. Fill in each box with a number and each circle with an operation to rewrite $R(x)$ in factored form.

$$R(x) = (\boxed{}\,x \bigcirc \boxed{})(\boxed{}\,x \bigcirc \boxed{})$$

2. Two graphs of $R(x)$ are shown.



a. Which key feature(s) shown on the graph can be determined from the factored form of $R(x)$?

b. Which key feature(s) shown on the graph can be determined from the standard form of $R(x)$?

c. Which key feature(s) shown on the graph cannot be determined from the factored or the standard form of $R(x)$?

8.9.2 Guided Instruction: Finding the Vertex Using Another Strategy

Sometimes completing the square is not an efficient method to find the vertex, or the process becomes cumbersome.

For example, the vertex for $R(x)$ in the Warm-Up is $(2.875, 3.0625)$. On the graph in the Warm-Up, the graphing software rounded the y-value to the nearest thousandth.

The steps of rewriting $R(x) = 4x^2 - 23x + 30$ to vertex form are shown.

$$R(x) = 4\left(x^2 - \frac{23}{4}x\right) + 30$$

$$R(x) = 4\left(x^2 - \frac{23}{4}x + \frac{529}{64}\right) + 30 - \frac{529}{16}$$

$$R(x) = 4\left(x - \frac{23}{8}\right)^2 - \frac{49}{16}$$

1. Discuss with your partner how the a and b coefficients of $R(x) = 4x^2 - 23x + 30$ relate to the value of h, $\frac{23}{8}$, of the vertex form of $R(x)$.

The y-value of the vertex in the function, $R(x)$, can be found by substituting the x-value of the vertex into the quadratic equation.

2. During the first year of production for the startup company in the Warm-Up, when will the company's revenue be negative?

3. Does it make sense for a company that's just starting to have a negative revenue?

8.9.3 Collaboration: Interpreting Key Features

Work with your partner to complete the following.

- Determine the key feature that is needed to answer the question.
- Identify the best form to use to find the key feature of the quadratic.
- Determine the answer and show your work.

1. A scientist is performing a study on mosquitoes. She collects water with mosquito larvae and places it in an enclosed container in her lab to watch for the larvae to hatch. The scientist uses the data from the study to create the function $M(t) = -3t^2 + 57t - 144$, where t is the number of days in the study and M is number of mosquitos in the container.

   a. When did mosquitoes first appear in the container?

Key Feature Needed	Which Form Reveals This?	Answer

   b. When were there are no more mosquitos in the container?

Key Feature Needed	Which Form Reveals This?	Answer

2. The function $C(x) = -2x^2 + 38x + 40$ models the sales, in hundreds of millions of dollars, of compact discs in the years since 1990.

   a. How many years did the sales of compact discs increase?

Key Feature Needed	Which Form Reveals This?	Answer

   b. When were the sales of compact discs at their greatest?

Key Feature Needed	Which Form Reveals This?	Answer

   c. What was the maximum amount of sales, in millions of dollars, of compact discs?

Key Feature Needed	Which Form Reveals This?	Answer

   d. Compact discs, or CDs, were the most popular way to listen to music before music streaming became available. Discuss with your partner why the key features of the function C make sense given this context.

8.9.4 Your Turn!

1. Engineers are designing a rollercoaster that goes underground at one point. The rollercoaster's path underground is a large curve. Engineers use the function shown to model the elevation, E, in feet and time, t, in seconds, of the rollercoaster's path for a portion of the ride.

$$E(t) = t^2 - 10t + 16$$

   a. Write an equivalent form of $E(t)$ that can be used to find the time when the rollercoaster goes underground and the time the rollercoaster returns above ground.

   b. When did the rollercoaster go underground?

   c. How long was the rollercoaster underground?

8.9.5 Wrap-Up: Ticket Out the Door

1. The equation $H(x) = -16x^2 + 96x + 6$ models the height, H, in feet, of a toy rocket x seconds after it is launched from a launchpad 6 feet above the ground.

   Complete the table and show your work to determine when the rocket is at its greatest height.

Key Feature Needed	Which Form Reveals This?	Answer